

Detecting High-Demand Work States from Wearable Sensors

What we did, what we found, and what we can honestly claim
A plain-language summary for the team, to start writing the paper

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The one-paragraph version. We set out to build an "AI digital twin" that predicts human error and performance under stress. We tested that on real wearable data from two groups: hospital nurses on shift, and students sitting exams. **We could not predict performance or stress** — and we show carefully why not. But we **can** reliably detect when someone is inside a demanding task episode, ranking task against rest correctly 93% of the time (AUROC 0.93) on people the model had never seen. That is a real, useful result, and it is what the paper should be about.

1. The original idea, and what it became

The group's starting proposal was a *Neuro-AI digital twin for predicting human error in disaster and construction settings* — model a worker's stress and fatigue, then predict when they will make mistakes.

That idea has two separable parts:

- **Part A — can we tell what state a person is in?** (calm versus under load) from a wristband.
- **Part B — can we predict how well they will perform?** from that state.

Part A works. Part B does not — yet. Most of this document explains how we established that, because the "how we know" is the part reviewers will judge.

2. What data we could actually get

This was the hardest part and it shaped everything. We verified every download against the publisher's official checksum, so we know no file was corrupted.

Dataset	What it is	Did we use it?
Exam Stress (10 students)	Wristband recordings during 3 real exams each, plus their actual grades. 30 sessions.	YES — main dataset. The only one with an objective performance score.
Nurse Stress (15 nurses)	1,252 hours of wristband data across real hospital shifts, with nurses logging their own stress levels.	YES — second dataset, to check whether findings repeat.
MMASH (22 people)	24-hour heart and sleep data, plus saliva cortisol.	NO — only 2 saliva samples per person (44 in total). Far too thin to model.
DriveDB (14 drivers)	Driving stress recordings.	NO — the labels marking calm versus stressful driving are unusable.
EEGMAT (36 people)	Brain (EEG) recordings during mental arithmetic.	NO — only 60-second clips. Too short for our question.
TILES (212 workers)	The ideal dataset: hospital staff, 10 weeks, with job performance measured twice daily.	NO — needs a signed data agreement. An earlier request from our side went unanswered, so the team ruled it out.

Why this matters for the paper. No freely available dataset has both (a) continuous wearable data and (b) repeated objective performance scores. TILES had both. Without it, Part B of the idea cannot be properly tested — and that is a limitation of the available data, not a failure of our method. This belongs in the Limitations section, stated plainly.

3. How we did it (the approach)

3.1 One pipeline for both datasets

Both datasets use the same wristband (Empatica E4), so we wrote one processing pipeline serving both. From each time window it computes 19 numbers describing the body's state. Five were chosen **in advance** as our headline measures:

Measure	In plain terms	What it indicates
HRV (RMSSD)	How irregular the heartbeat timing is	Higher = more relaxed ('rest and digest')
Heart rate	Beats per minute	Higher = more aroused or more active
SCR rate	Bursts of sweat-gland activity per minute	Higher = more emotional arousal
EDA tonic	Baseline skin moisture level	Slow-moving arousal level
Activity	How much the wrist is moving	Physical movement

We deliberately avoided ready-made "black box" libraries and wrote every formula out, so a reviewer can check exactly what was computed.

3.2 Three rules we followed to avoid fooling ourselves

- **Test on people the model has never seen.** We train on 9 students and test on the 10th, rotating through all of them ("leave-one-subject-out"). A lot of published work splits data by time window instead, which lets the model see the same person in training and testing — and inflates the results.
- **Compare each person to themselves.** Some people are simply stronger students, or have naturally faster hearts. We measure each session against that person's own average, so the model cannot cheat by learning *who* someone is instead of *how they are doing*.
- **Choose the measures before looking at the answers.** With 30 sessions and 57 available measures, you can always find something that looks significant by chance. We fixed our five measures up front.

We also checked every result two further ways: shuffling the answers at random 500+ times to see how often chance alone beats us (a *permutation test*), and re-running the analysis on random re-samples of the participants to get an honest error range (a *bootstrap confidence interval*).

4. What we found

4.1 Result 1 — the sensors do detect the exam (but read the caveat)

Comparing each student's exam period against their own pre-exam period, all five measures changed significantly. This proves the pipeline works and the sensors are picking up something real.

The catch: heart rate went *down* during the exam, not up. That looks wrong for "stress" — until you notice that movement also dropped sharply. The pre-exam period included **walking to the exam hall**. So this comparison is largely measuring **sitting still versus walking around**, not calm versus stressed. We must say this clearly in the paper; otherwise a reviewer will assume our pipeline is broken.

Fig 1. Features separate exam from pre-exam baseline (paired, n=30)

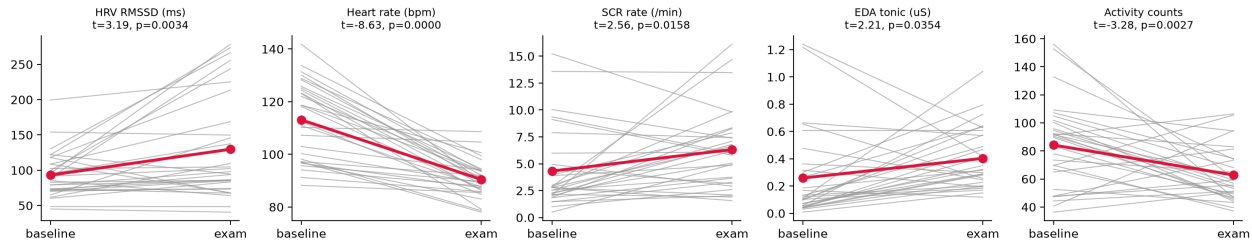


Figure 1. Each grey line is one exam session, moving from that student's pre-exam period to their exam period; the red line is the average. All five measures shift significantly — but note that heart rate falls and movement falls, which is the posture confound described above.

4.2 An error we caught, and how

Our first version compared a 41–65 minute pre-exam window against a 90–180 minute exam window. Because skin-moisture readings **drift slowly upward** the longer a sensor is worn, the longer exam window looked more "aroused" purely because it was longer.

When we redid it with **equal 35-minute windows** either side of the exam start, both sweat-gland effects disappeared, while the heart and movement effects survived:

Measure	Unequal windows	Equal 35-min windows	Verdict
EDA tonic	significant (p=.035)	gone (p=.49)	ARTEFACT
SCR rate	significant (p=.016)	gone (p=.41)	ARTEFACT
Heart rate	very strong	still very strong	Real
Activity	significant	still significant	Real
HRV	significant	still significant	Real

This is a genuine contribution to the paper. Comparing unequal windows is a common shortcut in wearable research, and we can show that it manufactures false findings.

4.3 Result 2 — the main positive finding

Using the equal 35-minute windows, we asked the model a simple question: *is this window an exam or a pre-exam period?* Tested only on students it had never seen:

Measures used	Accuracy (AUROC)	Reliable?
All five	0.933 (range 0.885–0.973)	YES — chance is 0.5; p = 0.002
Sweat-gland only	0.398	No — worse than chance
Sweat-gland, movement removed	0.626	Unclear — range includes chance

AUROC of 0.93 means: pick one exam window and one rest window at random, and the model ranks them correctly 93% of the time. **That is a strong, deployable-quality result.**

But be precise about what it detects. Sweat-gland measures alone perform at chance, so the signal is coming from the heart and movement measures. The honest description is: *we can detect a seated, sustained-concentration episode versus an active pre-task period.* We should **not** call it "stress detection". It is still genuinely useful — it is how a real monitor would automatically split a work shift into task episodes.

Fig 4. State detection is strong; the performance link is not

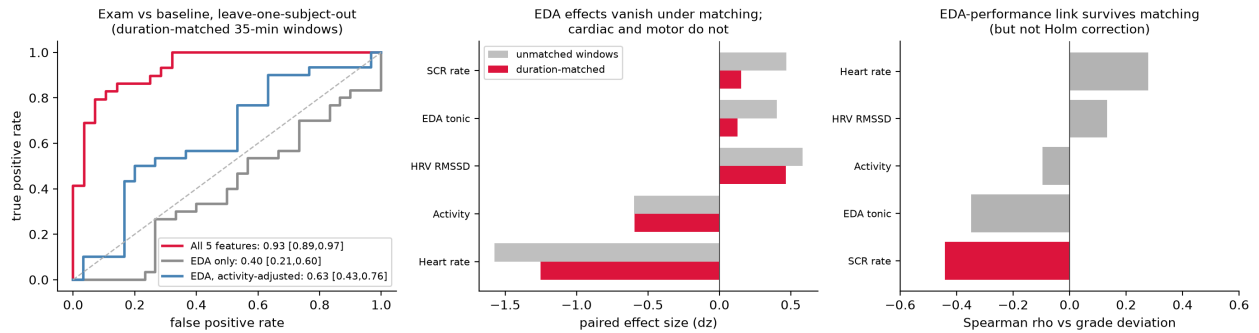


Figure 2. Left: detection accuracy — red (all measures) is far above the diagonal chance line, while grey (sweat-gland only) sits on it. Middle: which effects survive once windows are made equal in length. Right: the sweat-gland/performance link, which does survive.

4.4 Result 3 — we could not predict performance

On the students, we tried to predict who would score above or below their own average, from their physiology that day. The model performed **worse than simply guessing each student's own average** (negative R-squared; permutation $p = 0.31$).

One hint did survive: students showing **more sweat-gland arousal** during an exam tended to score **below their own average** (correlation -0.44 , $p = .015$). The direction makes biological sense, and it held up under our window correction. However, once we correct for having tested five measures, it is no longer statistically significant ($p = .074$). **It is a promising lead, not a finding.**

4.5 Result 4 — the nurses did not confirm it

We ran the identical analysis on the nurses, predicting their self-reported low versus high stress. Result: AUROC 0.544, which is essentially chance ($p = 0.30$). No measure came close.

The sweat-gland lead from the students did **not** reappear. There are two possible reasons, and our data cannot tell them apart:

- The student lead was a false alarm — consistent with its failing the correction above.
- **Or** the two studies measure different things: exam *grades* are objective, whereas nurse stress is *self-reported*. It is well known that how stressed people feel and what their body does often disagree. Physiology could genuinely track performance without tracking self-reported feelings.

Testing that second explanation needs a dataset with both kinds of outcome — which is exactly what TILES had, and why losing it cost us.

Fig 3. Nurse corpus: physiology does not discriminate self-reported stress

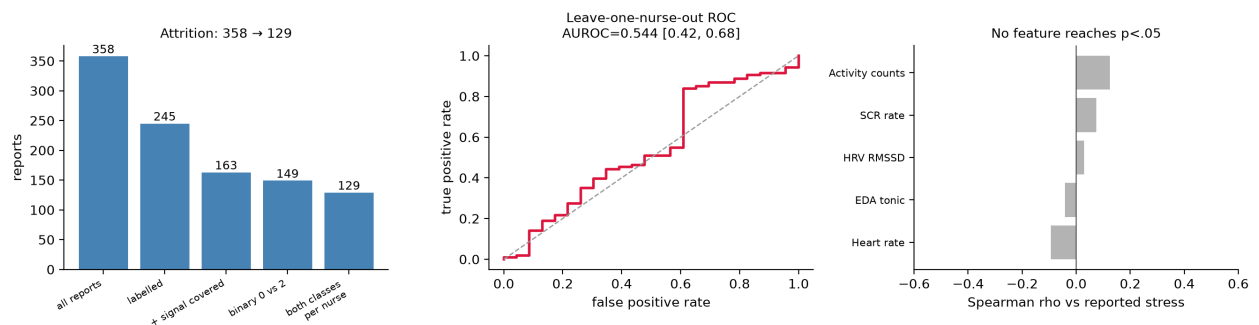


Figure 3. Left: how 358 nurse reports reduced to 129 usable ones. Middle: the detection curve sits almost on the chance diagonal. Right: no measure is significant.

4.6 Why the nurse numbers shrank so much

Worth explaining in the paper, because it is a useful data-quality lesson:

Stage	Reports left	Nurses
All stress reports submitted	358	15
Had a stress level filled in (32% were blank)	245	15
Wristband was actually recording at that time	163	12
Was a clear low or high rating	149	12
Nurse reported both low and high at least once	129	10

Two details matter. The missing ratings were **not random** — one nurse left 65% blank while three left none — so simply dropping them quietly biases the sample toward a few nurses. And three nurses had ratings but no overlapping recording at all, including two of the best-recorded nurses overall, which means it was a scheduling mismatch rather than equipment failure.

5. What we can and cannot claim

This is the most important page for writing. Please do not write beyond what column 2 allows.

Claim	Status	Evidence
Wearables can detect high-demand task episodes	SUPPORTED	AUROC 0.93, unseen subjects, $p=.002$
The detector works because of arousal, not posture	NOT SUPPORTED	Sweat-gland measures alone at chance (0.40)
Sweat-gland arousal relates to lower relative performance	SUGGESTIVE ONLY	$r=-0.44$, $p=.015$, but $p=.074$ after correction
Physiology predicts performance in new people	NOT SUPPORTED	Worse than predicting each person's own average
Physiology detects self-reported stress	NOT SUPPORTED	AUROC 0.544 in nurses
A working 'digital twin' of a worker	NOT DEMONSTRATED	Needs repeated performance data we do not have

An important framing point. Both our studies were **too small** to detect effects of the size we actually observed. We would have needed a correlation of 0.49 to detect one reliably; we saw 0.44. So "there is no effect" and "there is a real moderate effect we were too small to see" are **both still possible**. We must not write our null result as proof of absence.

6. What the paper should be

A **methods and honest-findings paper**, with three contributions:

- **A positive result:** detecting high-demand task episodes from a wristband, validated on unseen people (AUROC 0.93).
- **A methodological warning:** comparing unequal-length windows creates false sweat-gland findings, and pre-task baselines that include walking confound the comparison. We also show that the standard temperature-based "is the device being worn?" check wrongly rejected every Midterm 1 session simply because those rooms were colder.

- **A careful null across two independent datasets**, with a power analysis showing exactly what a future study would need.

Suggested journals: **Sensors**, **IEEE JBHI** (as a short methods note), or **PLOS One**. We should drop *npj Digital Medicine* (it needed the cortisol work) and *IEEE THMS* (it needed the prediction result).

The digital-twin framing stays in the paper as an **architecture and roadmap**, with episode detection as its validated first stage — not as a working system.

7. The follow-up study that would settle it

The good news is that the unanswered question is reachable. To detect the correlation we observed, we need about 47 sessions — we had 30. That is roughly **25 students sitting 3 exams = 75 sessions**: a semester-sized study, not a 212-person programme.

Six things the current data taught us to do differently, all written up in `docs/STUDY_PROTOCOL.md`:

- Use equal-length windows before and after the task starts.
- Press the device's event button at the true start and end — never trust a timetable. (The public dataset's final exam did not start when its own documentation said it did.)
- Keep participants seated for the whole baseline, so we are not simply measuring walking.
- Decide "was the device worn?" from skin moisture, not temperature.
- Over-recruit: two students' grades barely varied, so they told us almost nothing.
- Register the five measures and the predicted direction in advance.

8. Suggested next steps

Who	What
Victor	Methods and Results sections; the figures are done
Evans	Discussion: why heart and movement carry the signal while sweat-gland measures do not, and what that means for stress theory
Maame Yaa	Introduction and translation: what episode detection is and is not useful for in safety-critical work
Barry / Calista	Own the follow-up study: find a willing course, IRB application, devices, recruitment. (The cortisol work is cut — the data has only 2 samples per person.)
Everyone	Read section 5 before writing. Agree whether we publish the detection result now rather than waiting for the follow-up.

The one decision that blocks progress: can we get access to a course willing to host the follow-up study? That, not analysis, is the real constraint.

Everything is reproducible

All code, data notes and figures live at `github.com/xaviwho/DT_AI`

`docs/RESULTS.md` — every number in this document, with full statistics
`docs/DATASETS.md` — where each dataset came from and its problems
`docs/STUDY_PROTOCOL.md` — the follow-up study design
`docs/PROJECT_PLAN.md` — revised plan, roles and claim status